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Activity of cefepime/zidebactam against MDR *Escherichia coli* isolates harbouring a novel mechanism of resistance based on four-amino-acid inserts in PBP3

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Background: Recent reports reveal the emergence of *Escherichia coli* isolates harbouring a novel resistance mechanism based on four-amino-acid inserts in PBP3. These organisms concomitantly expressed ESBLs or/and serine-/metallo-carbapenemases and were phenotypically detected by elevated aztreonam/avibactam MICs.

Objectives: The *in vitro* activities of the investigational antibiotic cefepime/zidebactam and approved antibiotics (ceftazidime/avibactam, ceftolozane/tazobactam, imipenem/relebactam and others) were determined against *E. coli* isolates harbouring four-amino-acid inserts in PBP3.

Methods: Whole-genome sequenced *E. coli* isolates ($n = 89$) collected from a large tertiary care hospital in Southern India ($n = 64$) and from 12 tertiary care hospitals located across India ($n = 25$) during 2016–18, showing aztreonam/avibactam MICs ≥ 1 mg/L (≥ 4 times the aztreonam epidemiological cut-off) were included in this study. The MICs of antibiotics were determined using the reference broth microdilution method.

Results: Four-amino-acid inserts [YRIK ($n = 30$) and YRIN ($n = 53$)] were found in 83/89 isolates. Among 83 isolates, 65 carried carbapenemase genes [bla_{NDM} ($n = 39$), $bla_{OXA\text{-}48\text{-}like}$ ($n = 11$) and $bla_{NDM} + bla_{OXA\text{-}48\text{-}like}$ ($n = 15$)] and 18 isolates produced ESBLs/class C β-lactamases only. At least 16 unique STs were noted. Cefepime/zidebactam demonstrated potent activity, with all isolates inhibited at ≤ 1 mg/L. Comparator antibiotics including ceftazidime/avibactam and imipenem/relebactam showed limited activities.

Conclusions: *E. coli* isolates concurrently harbouring four-amino-acid inserts in PBP3 and NDM are an emerging therapeutic challenge. Assisted by the PBP2-binding action of zidebactam, the cefepime/zidebactam combination overcomes both target modification (PBP3 insert)- and carbapenemase (NDM)-mediated resistance mechanisms in *E. coli*.

Introduction

Gram-negative bacteria display resistance to β-lactams, assisted by a triple line of defences. The double-membrane wall acts as the first line of defence by restricting the permeation of antibiotics, while the second line of defence comprising periplasmic β-lactamases and efflux pumps further impedes the binding to their target PBP. Even if a β-lactam evades the first two lines of defence, modifications in the PBPs may act as the third line of defence, rendering the drug less effective. While β-lactamases and efflux pumps are ubiquitously expressed in diverse Gram-negative bacteria, until recently, PBP modifications have been largely restricted to *Pseudomonas aeruginosa*.[1] However, recent reports from several countries show the presence of novel four-amino-acid inserts in *Escherichia coli* PBP3, which is a primary target for many β-lactams.[2–5] This novel resistance mechanism is found in both carbapenemase- and ESBL-producing *E. coli* and is phenotypically identified based on raised aztreonam/avibactam MICs.

After the initial reports from Alm *et al.*[2] and Zhang *et al.*[3] we recently assessed the prevalence of the PBP3 insert-based resistance mechanism in India.[5] We found that 23% of *E. coli* expressing ESBL or/and OXA-48-like/MBL showed elevated aztreonam/avibactam MICs (2–32 mg/L), with 72/76 such isolates harbouring YRIN or

YRIK insertions in PBP3. The large-scale presence of this resistance mechanism in *E. coli* is worrying as it is the most common pathogen in hospital/community infections. Moreover, identification of this resistance mechanism in a routine diagnostic set-up is challenging.

In light of several reports of *E. coli* isolates concurrently harbouring PBP3 inserts and diverse β-lactamases, it is important to evaluate the activities of newer drugs against such isolates. Conceivably, PBP3 insert-mediated resistance could be overcome by multiple PBP-binding carbapenems; however, concomitantly expressed carbapenemases compromise their efficacy. Thus, altered PBP3 combined with expression of carbapenemases in *E. coli* represents an emerging clinical threat.

WCK 5222 is a combination of cefepime and a novel β-lactam enhancer, zidebactam, slated to enter Phase 3 development. Zidebactam, a non-β-lactam bicycloacyl hydrazide pharmacophore, shows high-affinity PBP2 binding.[6] Past studies established cefepime/zidebactam's activity against serine-/metallo-carbapenemase-producing Gram-negative isolates with concurrent expression of hyper-efflux and impermeability.[7–11] To the best of our knowledge, this is the first study evaluating cefepime/zidebactam activity in comparison with several other antibiotics against *E. coli* isolates showing dual resistance (PBP3 modification and ESBL/MBL/OXA-48-like expression).

Materials and methods

Bacterial isolates

E. coli isolates ($n = 64$) showing aztreonam/avibactam MICs ≥1 mg/L (≥4 times the aztreonam epidemiological cut-off value) were identified among the blood culture isolates collected during 2017–18 at Christian Medical College Hospital, Vellore, India. Additionally, 25 *E. coli* isolates (aztreonam/avibactam MICs ≥1 mg/L) collected from 12 other Indian tertiary hospitals (minimum of 2 isolates from each hospital) in 2016–18 were included.[6] These isolates were from clinical samples, such as pus, wound swabs, sputum, tracheal secretions, blood and urine.

MIC determination

The MICs of cefepime/zidebactam (1:1 ratio) and comparators were determined using the broth microdilution method (CLSI, M07-A11).[12] Commercial preparations with >95% HPLC purity of cefepime, ceftazidime, imipenem, meropenem, aztreonam, ceftriaxone, gentamicin, amikacin and levofloxacin were used, while piperacillin and cefoperazone were from Sigma–Aldrich, USA. Zidebactam, avibactam and relebactam were synthesized in-house (>95% purity); tazobactam was procured from Qilu Tianhe Pharmaceutical Co. Ltd (China). The quality control strains, *E. coli* ATCC 25922, *Acinetobacter baumannii* NCTC 13304 and *P. aeruginosa* ATCC 27853, were included in each run. The susceptibilities were interpreted based on EUCAST criteria.

WGS

Genomic DNA from *E. coli* was extracted using a QIAamp DNA mini kit (QIAGEN, Germany). Sequence libraries were constructed employing an Illumina Nextera library preparation protocol (Illumina, Inc., San Diego, CA, USA). WGS was performed on an Illumina HiSeq platform. The raw reads were paired, trimmed and assembled *de novo* by Unicycler v0.4.8.[13] Genome annotation was performed using Prokka v1.14.6[14] and the NCBI prokaryotic genome automatic annotation pipeline (http://www.ncbi.nlm.nih.gov/genomes/static/Pipeline.html).[15] Further, ResFinder and MLST 2.0 from the CGE server (http://www.cbs.dtu.dk/services) were used for antimicrobial resistance and epidemiological analysis. The *ftsI* sequences from *E. coli* were retrieved employing a local BLAST algorithm with reference WT sequence (*E. coli* NCTC 9022; accession no. LR134237). The sequences of clinical isolates were aligned with WT sequence (Clustal Omega, https://www.ebi.ac.uk/Tools/msa/clustalo/). Alignments were visualized using SeaView v4.[16]

Results

Analysis of *ftsI* sequences revealed the presence of four-amino-acid insertions in PBP3 after residue 333 in 83/89 *E. coli* isolates exhibiting elevated aztreonam/avibactam MICs of ≥1 mg/L. Consistent with previous observations,[2,5] two types of inserts were noted: YRIK ($n = 30$) and YRIN ($n = 53$). Carbapenemases were detected in 65/83 isolates [bla_{NDM} ($n = 39$), $bla_{\text{oxa-48-like}}$ ($n = 11$) and bla_{NDM} + $bla_{\text{oxa-48-like}}$ ($n = 15$)] carrying PBP3 inserts and the remainder harboured ESBLs/class C β-lactamases. Apart from carbapenemase genes, bla_{CMY} (51/83), $bla_{\text{CTX-M-15}}$ (41/83) and $bla_{\text{OXA-1}}$ (37/83) were frequently encountered. MLST profiling showed the existence of at least 16 unique STs (Table S1, available as Supplementary data at *JAC* Online).

The impact of four-amino-acid inserts in PBP3 on aztreonam/avibactam was evident from its elevated MICs (1 to >32 mg/L), wherein higher MICs were observed for YRIK inserts compared with YRIN inserts (geometric mean 7.8 versus 3.7 mg/L, data not shown). This observation is in agreement with our previous study.[5]

Table 1. *In vitro* activities (MIC_{50} and MIC_{90}) of cefepime/zidebactam and other antibiotics against 89 *E. coli* isolates exhibiting elevated aztreonam/avibactam MICs of ≥1 mg/L

	MIC (mg/L)			
Antibacterial agent	range	MIC_{50}	MIC_{90}	Percentage susceptible[b]
Cefepime	8 to >32	>32	>32	0
Cefepime/zidebactam[a]	≤0.06–1	0.12	0.5	100[c]
Aztreonam/avibactam	1 to >32	4	16	12.35[d]
Piperacillin/tazobactam	8 to >32	>64	>64	2.24
Cefoperazone/sulbactam	4 to >32	>32	>32	NA
Meropenem	≤0.06 to >8	>8	>8	37.07
Ceftolozane/tazobactam	4 to >32	>32	>32	0
Ceftazidime	2 to >16	>16	>16	0
Ceftazidime/avibactam	≤0.06 to >32	>32	>32	34.83
Imipenem/relebactam	≤0.06 to >32	16	32	34.83
Ceftriaxone	1 to >2	>2	>2	1.12
Gentamicin	0.25 to >8	>8	>8	40.44
Amikacin	1 to >32	4	>32	56.18
Levofloxacin	0.06 to >4	>4	>4	1.12

NA, not applicable.
[a]MICs of cefepime/zidebactam were determined at a 1:1 ratio.
[b]Susceptiblity was interpreted based on EUCAST criteria.
[c]Susceptiblity was determined based on EUCAST interpretive criteria for cefepime.
[d]Susceptiblity was determined based on EUCAST interpretive criteria for aztreonam.

Table 2. MIC distributions of cefepime, cefepime/zidebactam and various comparators for *E. coli* isolates harbouring four-amino-acid inserts in PBP3 (stratified based on β-lactamases)

β-Lactamase (number of isolates)/antibiotic	No. of isolates inhibited at respective MIC (mg/L) (% cumulative inhibition)										
	≤0.06	0.12	0.25	0.5	1	2	4	8	16	32	>32
All *E. coli* (*n* = 89)											
cefepime								6 (6.7)	6 (13.5)	6 (20.2)	71 (100)
cefepime/zidebactam	8 (10.1)	41 (55.5)	27 (85.4)	11 (97.7)	2 (100.0)						
aztreonam/avibactam					11 (12.3)	16 (30.3)	28 (61.8)	16 (79.7)	12 (93.2)	5 (98.9)	1 (100)
meropenem	17 (19.1)	8 (28.1)	0 (28.2)	1 (29.2)	5 (34.8)	2 (37.0)	1 (38.2)	2 (40.4)	53[a] (100.0)		
ceftazidime/avibactam	2 (2.2)	0 (2.2)	0 (2.2)	1 (3.4)	2 (5.6)	10 (16.9)	13 (31.5)	3 (34.8)	0 (34.8)	0 (34.8)	58 (100)
imipenem/relebactam	4 (4.5)	8 (13.5)	11 (25.8)	3 (29.2)	4 (33.7)	1 (34.8)	1 (35.9)	7 (43.8)	29 (76.4)	12 (89.9)	9 (100)
ESBL/class C (*n* = 18)											
cefepime								1 (5.6)	2 (16.7)	3 (33.3)	12 (100)
cefepime/zidebactam	2 (16.6)	7 (50.0)	6 (83.3)	3 (100.0)							
aztreonam/avibactam					2 (11.1)	3 (27.8)	5 (55.5)	3 (72.2)	4 (94.4)	1 (100.0)	
meropenem	10 (55.6)	4 (77.7)	0 (77.7)	0 (77.7)	1 (83.3)	0 (83.3)	1 (88.8)	0 (88.8)	2[a] (100.0)		
ceftazidime/avibactam	1 (5.5)	0 (5.5)	0 (5.5)	0 (5.5)	1 (11.1)	6 (44.4)	4 (66.6)	3 (83.3)	0 (83.3)	0 (83.3)	3 (100)
imipenem/relebactam	3 (16.7)	5 (44.4)	3 (61.1)	3 (77.8)	1 (83.3)	0 (83.3)	1 (88.9)	0 (88.9)	2 (100)		
NDM (*n* = 39)											
cefepime											39 (100)
cefepime/zidebactam	3 (7.7)	22 (64.1)	10 (89.7)	3 (97.4)	1 (100.0)						
aztreonam/avibactam					6 (15.4)	6 (30.8)	12 (61.5)	8 (82.0)	5 (94.9)	1 (97.4)	1 (100)
meropenem					1 (2.5)	1 (5.1)	0 (5.1)	2 (10.2)	35[a] (100.0)		
ceftazidime/avibactam					1 (2.5)	0 (2.5)	0 (2.5)	0 (2.5)	0 (2.5)	0 (2.5)	38 (100)
imipenem/relebactam					1 (2.5)	0 (2.5)	0 (2.5)	7 (20.5)	15 (58.9)	10 (84.6)	6 (100)
OXA-48-like (*n* = 11)											
cefepime								4 (36.4)	3 (63.6)	1 (72.7)	3 (100)
cefepime/zidebactam		4 (36.3)	6 (90.9)	1 (100.0)							
aztreonam/avibactam						3 (27.3)	3 (54.5)	2 (72.7)	2 (90.9)	1 (100)	
meropenem	4 (36.4)	3 (63.6)	0 (63.6)	1 (72.7)	3 (100.0)						
ceftazidime/avibactam	1 (9.1)	0 (9.1)	0 (9.1)	1 (18.2)	0 (18.2)	3 (45.4)	5 (90.9)	0 (90.9)	0 (90.9)	0 (90.9)	1 (100)
imipenem/relebactam	2 (18.2)	2 (36.4)	4 (72.2)	0 (72.2)	2 (90.9)	1 (100)					
NDM + OXA-48-like (*n* = 15)											
cefepime											15 (100)
cefepime/zidebactam	2 (13.3)	5 (46.6)	4 (73.3)	4 (100.0)							
aztreonam/avibactam					2 (13.3)	2 (26.6)	6 (66.6)	3 (86.6)	0 (86.6)	2 (100)	
meropenem									15 (100.0)		
ceftazidime/avibactam											15 (100)
imipenem/relebactam									11 (73.3)	1 (80.0)	3 (100)

[a]MIC is greater than or equal to the indicated value.
NDM subtypes: NDM-5, NDM-7, NDM-4 and NDM-6 were in 50 isolates, 2 isolates, 1 isolate and 1 isolate, respectively.
OXA-48-like subtypes: OXA-232 and OXA-181 were in 14 and 12 isolates, respectively.

Based on EUCAST interpretive criteria for aztreonam, merely 12.4% isolates were susceptible to aztreonam/avibactam (Tables 1 and 2). Ceftazidime/avibactam MICs (1 to >32 mg/L, 86% were susceptible) were higher than its epidemiological cut-off (0.5 mg/L)[17] for 26/29 isolates carrying ESBL/class C/OXA-48-like enzymes and PBP3 inserts, indicating an adverse impact of the latter. Expectedly, ceftazidime/avibactam did not show activity against NDM-expressing and PBP3 insert-harbouring *E. coli*. Other comparators, such as piperacillin/tazobactam, cefoperazone/sulbactam and ceftolozane/tazobactam, were found to be almost inactive. Moreover, the activities of meropenem and imipenem/relebactam were limited, mainly due to a higher proportion of isolates expressing NDM/OXA-48-like carbapenemases. Imipenem/relebactam was active against the subset expressing ESBL/class C β-lactamases. Amikacin, gentamicin and levofloxacin showed poor activities, pointing to the MDR nature of these isolates.

The activity of cefepime/zidebactam (≤1 mg/L) was several fold higher than that of cefepime alone (MICs ≥8 mg/L) and cefepime/zidebactam inhibited all the isolates tested in this study at ≤1 mg/L (MIC_{90} 0.5 mg/L), irrespective of associated β-lactamases. Against the subset of *E. coli* isolates with NDM expression (n = 39) and PBP3 inserts, cefepime/zidebactam demonstrated potent activity (MIC range ≤0.06–1 mg/L).

Discussion

Though PBP modification is widespread in Gram-positive and certain fastidious Gram-negative pathogens, it remained a rare occurrence in Enterobacterales.[18] Therefore, emergence of *E. coli* isolates harbouring four-amino-acid insertions in PBP3 is a worrisome phenomenon. This novel resistance mechanism impacts aztreonam/avibactam, which is under development for infections caused by both serine- and metallo-carbapenemase-expressing Enterobacterales. In addition, the present study reveals the impact of this resistance mechanism on ceftazidime/avibactam as evidenced by the MIC shift observed in isolates harbouring PBP3 inserts and serine-β-lactamases. Ceftazidime/avibactam, meropenem and imipenem/relebactam were unable to cover the NDM-expressing subset of *E. coli* isolates. The extremely limited activity of ceftolozane/tazobactam and older β-lactams/β-lactamase inhibitors against serine-β-lactamase-expressing *E. coli* is possibly linked with multiple factors, such as enzymatic liability of β-lactams, their poor PBP3 binding affinities and inability of β-lactamase inhibitors to protect β-lactams from inactivation.

On the other hand, though cefepime also targets PBP3 of *E. coli*[6] and is vulnerable to ESBL and NDM, the cefepime/zidebactam combination retained potent activity against such *E. coli* (MIC_{100} 1 mg/L), irrespective of β-lactamases expressed and the presence of PBP3 inserts. This is ascribed to zidebactam's β-lactam-enhancer action. It appears that zidebactam's high-affinity PBP2 binding (IC_{50} 0.06 mg/L),[6] optimal permeation and universal stability to β-lactamases facilitate unhindered PBP2 binding, which compensates for the potential suboptimal binding of cefepime amidst carbapenemase expression and altered PBP3. Moreover, cefepime is helped by its secondary binding to PBP2 and PBP1a/b.[6] Though avibactam also possesses modest PBP2 inhibition (IC_{50} 0.92 mg/L)[19] and is stable to β-lactamases, it is not able to compensate for the loss in aztreonam or ceftazidime's binding to modified PBP3. It is particularly interesting to note that, despite aztreonam's stability to MBLs and avibactam's broad-spectrum β-lactamase inhibition feature, the combination is unable to overcome the resistance due to target modification. Therefore, rather than a combination of a β-lactamase inhibitor with an MBL-stable PBP3-targeting β-lactam, a combination of a PBP2 inhibitor and a PBP3-targeting β-lactam better overcomes the triple mechanism of resistance (serine-β-lactamases, MBLs and PBP3 inserts).

The bla_{NDM} gene was detected in 39/83 isolates that harboured PBP3 inserts and MLST profiling revealed the existence of diverse STs. A high proportion of bla_{NDM} in Indian *E. coli* isolates harbouring inserts in PBP3 was also previously noted, which could be due to a higher prevalence of NDM genes escaping among multiple ST lineages.[2] A recent report suggests that PBP3 insert clones emerge through recombination events and such isolates with *ompC* and *ompF* mutations preferentially acquire carbapenemase genes.[20]

In summary, we show for the first time (to the best of our knowledge), the potent activity of cefepime/zidebactam against *E. coli* isolates with novel four-amino-acid PBP3 inserts, irrespective of the types of β-lactamases co-expressed. Importantly, all the isolates studied were inhibited at ≤1 mg/L, a concentration lower than the cefepime EUCAST breakpoint. The PBP2-binding-driven β-lactam-enhancer mechanism of zidebactam enables the combination to tackle both enzymatic and non-enzymatic resistance mechanisms operating concurrently.

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Transparency declarations

S.S.B. and M.V.P. own shares of Wockhardt Ltd. All other authors: none to declare.

Supplementary data

Table S1 is available as Supplementary data at *JAC* Online.

References

1 Lopez-Causape C, Cabot G, Del Barrio-Tofino E *et al.* The versatile mutational resistome of *Pseudomonas aeruginosa*. *Front Microbiol* 2018; **9**: 685.

2 Alm RA, Johnstone MR, Lahiri SD. Characterization of *Escherichia coli* NDM isolates with decreased susceptibility to aztreonam/avibactam: role of a novel insertion in PBP3. *J Antimicrob Chemother* 2015; **70**: 1420–8.

3 Zhang Y, Kashikar A, Brown CA *et al.* Unusual *Escherichia coli* PBP 3 insertion sequence identified from a collection of carbapenem-resistant Enterobacteriaceae tested *in vitro* with a combination of ceftazidime-, ceftaroline-, or aztreonam-avibactam. *Antimicrob Agents Chemother* 2017; **61**: e00389-17.

4 Kazmierczak KM, Bradford PA, Stone GG *et al. In vitro* activity of ceftazidime-avibactam and aztreonam-avibactam against OXA-48-carrying Enterobacteriaceae isolated as part of the International Network for Optimal Resistance Monitoring (INFORM) global surveillance program from 2012 to 2015. *Antimicrob Agents Chemother* 2018; **62**: e00592-18.

5 Periasamy H, Joshi P, Palwe S *et al.* High prevalence of *Escherichia coli* clinical isolates in India harbouring four amino acid inserts in PBP3 adversely impacting activity of aztreonam/avibactam. *J Antimicrob Chemother* 2020; **75**: 1650–1.

6 Moya B, Barcelo IM, Udaykar A *et al.* WCK 5222 [cefepime-zidebactam, FEP-ZID]: mechanistic basis behind novel β-lactam–β-lactam enhancer combination against metallo-β-lactamase (MBL)-producing *E. coli* (EC) *K. pneumoniae* (KP), *P. aeruginosa* (PA) and its impact on therapeutically relevant bactericidal exposures assessed through *in vitro* pharmacodynamic modelling (IVPM) and mouse lung eradication studies. *IDWeek 2016, New Orleans, LA, USA*. Abstract 1982.

7 Moya B, Barcelo IM, Cabot G *et al. In vitro* and *in vivo* activities of β-lactams in combination with the novel β-lactam enhancers zidebactam and WCK 5153 against multidrug-resistant metallo-β-lactamase-producing *Klebsiella pneumoniae*. *Antimicrob Agents Chemother* 2019; **63**: e00128-19.

8 Avery LM, Abdelraouf K, Nicolau DP. Assessment of the *in vivo* efficacy of WCK 5222 (cefepime-zidebactam) against carbapenem-resistant

Acinetobacter baumannii in the neutropenic murine lung infection model. *Antimicrob Agents Chemother* 2018; **62**: e00948-18.

9 Avery LM, Mullane EM, Nicolau DP. Evaluation of the *in vitro* activity of WCK 5222 (cefepime/zidebactam) and currently available combination therapies against single- and double-carbapenemase producing Enterobacteriaceae: expanding the zone of hope. *Int J Antimicrob Agents* 2020; **55**: 105863.

10 Kidd JM, Abdelraouf K, Nicolau DP. Efficacy of human-simulated broncho-pulmonary exposures of cefepime, zidebactam and the combination (WCK 5222) against MDR *Pseudomonas aeruginosa* in a neutropenic murine pneumonia model. *J Antimicrob Chemother* 2020; **75**: 149–55.

11 Mullane EM, Avery LM, Nicolau DP. Comparative evaluation of the *in vitro* activities of WCK 5222 (cefepime-zidebactam) and combination antibiotic therapies against carbapenem-resistant *Pseudomonas aeruginosa*. *Antimicrob Agents Chemother* 2020; **64**: e01669-19.

12 CLSI. *Methods for Dilution Antimicrobial Susceptibility Tests for Bacteria That Grow Aerobically—Eleventh Edition*. 2018.

13 Wick RR, Judd LM, Gorrie CL *et al.* Unicycler: resolving bacterial genome assemblies from short and long sequencing reads. *PLoS Comput Biol* 2017; **13**: e1005595.

14 Seemann T. Prokka: rapid prokaryotic genome annotation. *Bioinformatics* 2014; **30**: 2068–9.

15 Tatusova T, DiCuccio M, Badretdin A *et al.* NCBI prokaryotic genome annotation pipeline. *Nucleic Acids Res* 2016; **44**: 6614–24.

16 Gouy M, Guindon S, Gascuel O. SeaView version 4: a multiplatform graphical user interface for sequence alignment and phylogenetic tree building. *Mol Biol Evol* 2010; **27**: 221–4.

17 Flamm RK, Stone GG, Sader HS *et al.* Avibactam reverts the ceftazidime MIC90 of European Gram-negative bacterial clinical isolates to the epidemiological cut-off value. *J Chemother* 2014; **26**: 333–8.

18 Zapun A, Contreras-Martel C, Vernet T. Penicillin-binding proteins and β-lactam resistance. *FEMS Microbiol Rev* 2008; **32**: 361–85.

19 Asli A, Brouillette E, Krause KM *et al.* Distinctive binding of avibactam to penicillin-binding proteins of Gram-negative and Gram-positive bacteria. *Antimicrob Agents Chemother* 2016; **60**: 752–6.

20 Patiño-Navarrete R, Rosinski-Chupin I, Cabanel N *et al.* Stepwise evolution and convergent recombination underlie the global dissemination of carbapenemase-producing *Escherichia coli*. *Genome Med* 2020; **12**: 10.